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North America 2025

SIG Multicluster Intro & Deep Dive

- What this SIG is about
- Roll Call of SIG Projects
- User Experience Survey Findings
- How to follow and contribute

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What this SIG is about

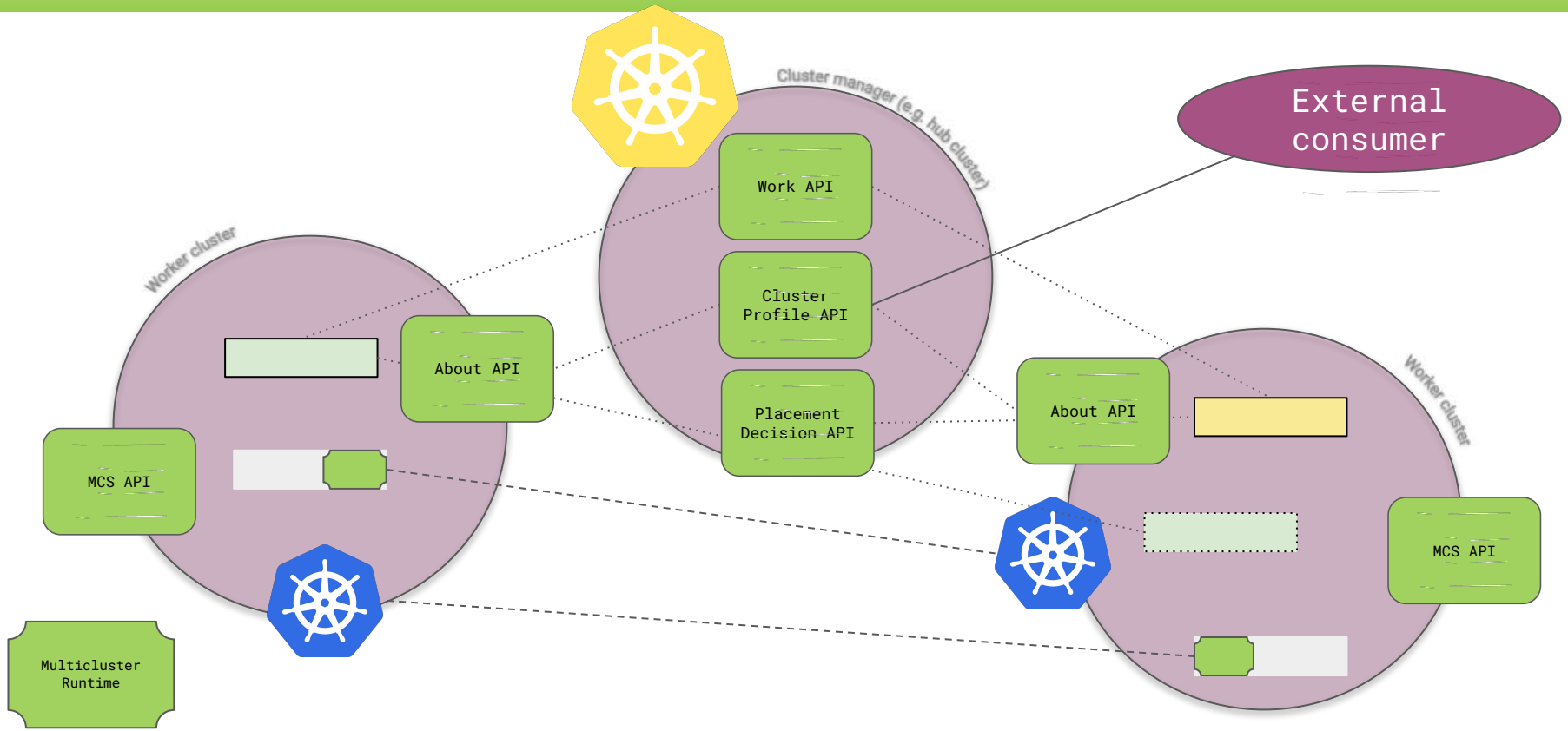
What this SIG is about

- Solving common challenges related to the management of multiple Kubernetes clusters, and applications that exist therein.
- Multi-cluster is everywhere
  - Environment isolation; Tenancy; Multi cloud; Fault tolerance; Data locality; Policy; Capacity; Performance
- Multi cluster ***applications*** (split across multiple clusters) can be powerful
- Kubernetes was built with the Cluster as the Universe
  - We’ve made some progress, but there’s a long way to go

Our approach

- Recognize diversity of the ecosystem
  - Where workloads run (cloud, on-prem)
  - Environmental constraints
  - Recognize the ecosystem when designing APIs – focus on APIs that work for everyone
  - Lift the standards existing or needed in the ecosystem
- Avoid solving any optional problems
  - “This could be useful” is an engineering fly trap!
- Consistency with existing APIs
- Composable building blocks

Project roll call!!!



User Experience Survey Findings

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Overview

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Multi Cluster Operations - User Research Survey

- **Goal:** Make complex systems easy to use, Turn difficult operations into simple human friendly.
- **Survey Design:** Engineers working with multi clusters | Problem areas: Environment, Tooling, observability, troubleshooting, automation, collaboration | Qualitative & Quantitative
- **Survey Reach:** Kubernetes SIG | 150 respondents
- **Research Data analysis:**
  Text analyzer = Written responses, Python = Numerical Data
  Demographics / Tools / Challenges
- **Roles (Demographics):** Platform (38%) | DevOps (26%) | SRE (10%) | Others (Dev & Cloud Eng.)

Qualitative highlights

Why organizations run multi clusters / Use case

Most common

Environment — Development, Staging, Production = **Test deployment safely, breaking live systems**

Region — Cluster/region = **high performance, Geo redundancy, better service availability**

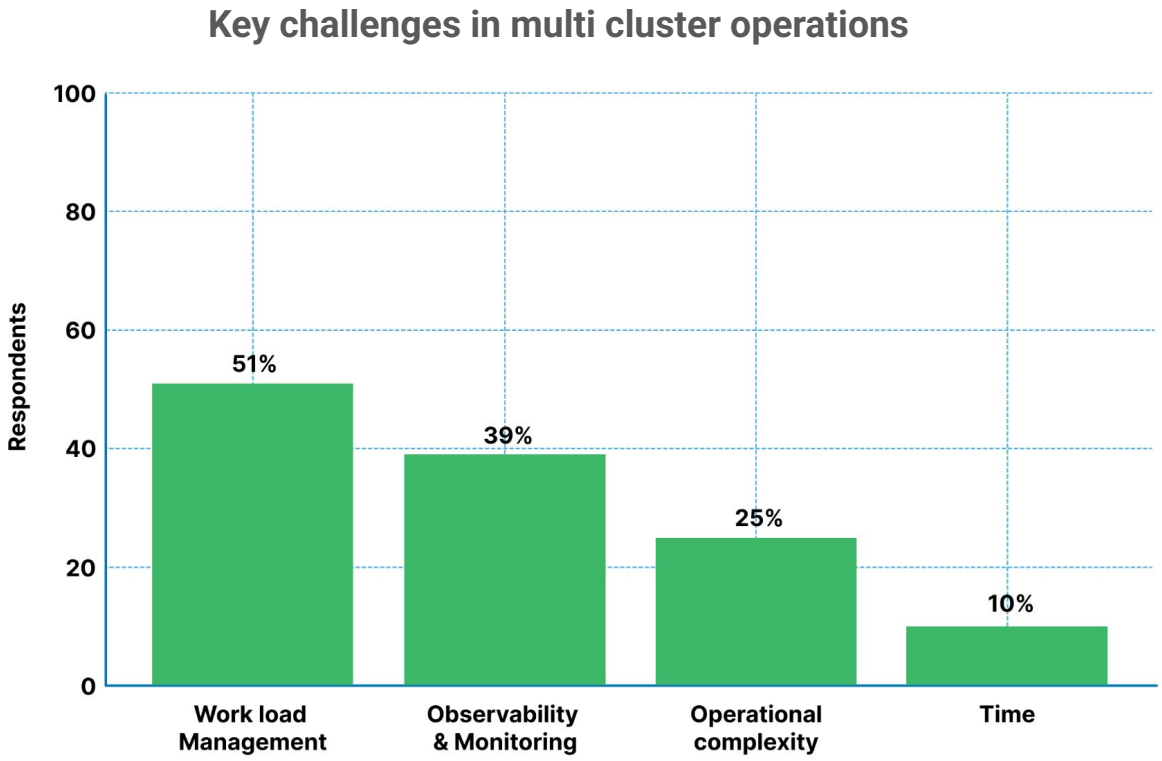
Business Unit — Cluster / business unit = **Team ownership, independence scaling and updates, easy compliance**

Isolation — Other isolation dimensions e.g. Cluster / Workload = **better security and management**

Infrastructure Provider — (Run clusters) Single or Multi cloud/ on prem/ Edge locations = **Flexible deployment, Independence from vendor lock-in**

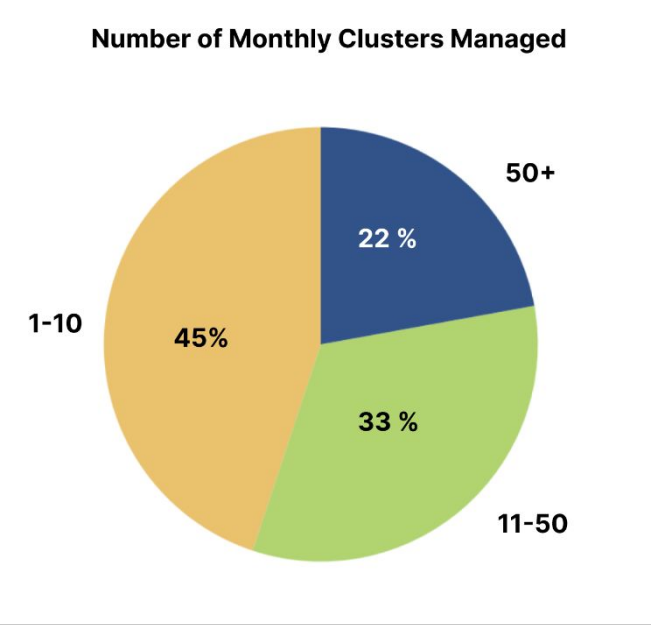
Common

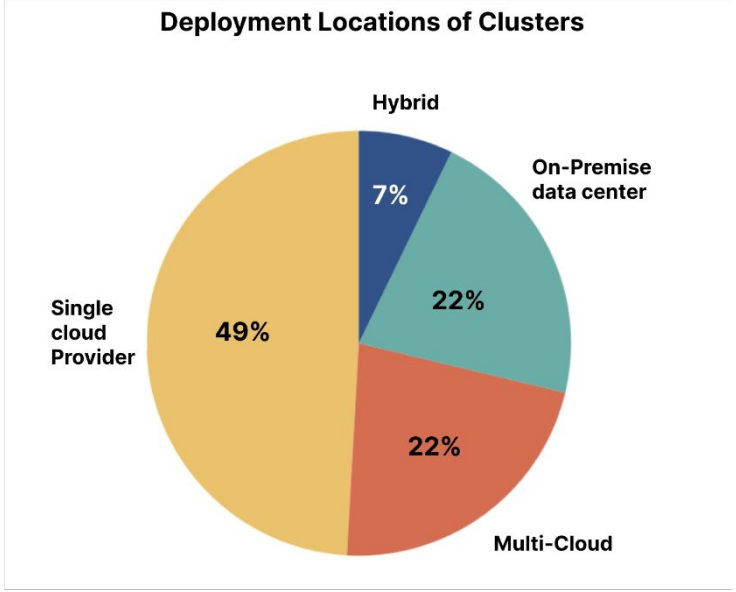
Qualitative highlights



- **What Engineers want:**
  - AI + Automation Ops
  - Flexible Architecture
  - Simplified UX
- **Unique Responses**
  - Dashboard as Code
  - Specialized hardware requirements

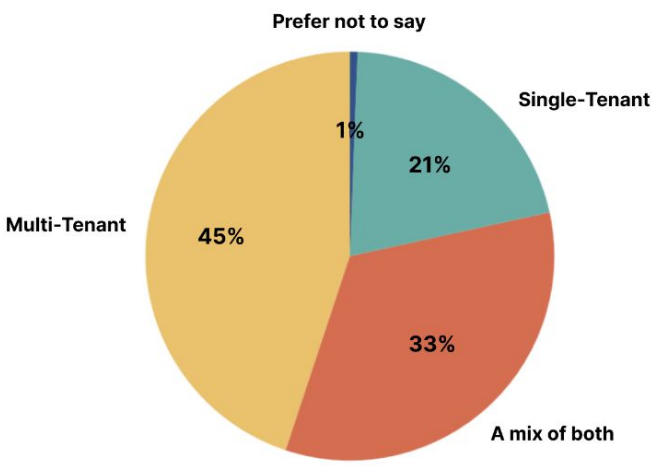
Quantitative highlights



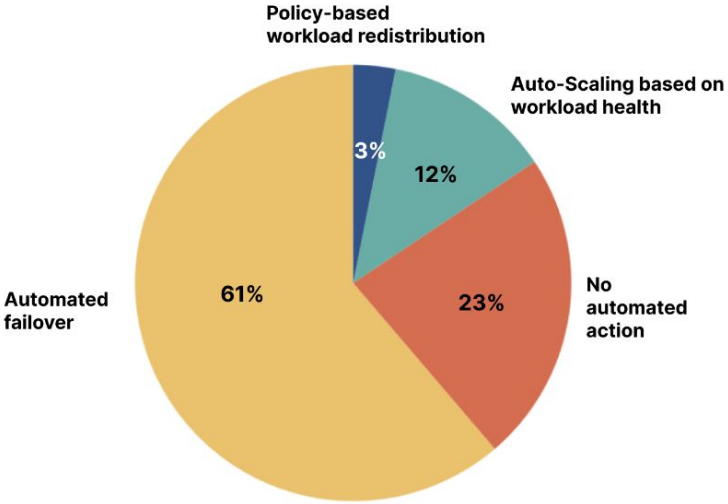


Quantitative highlights

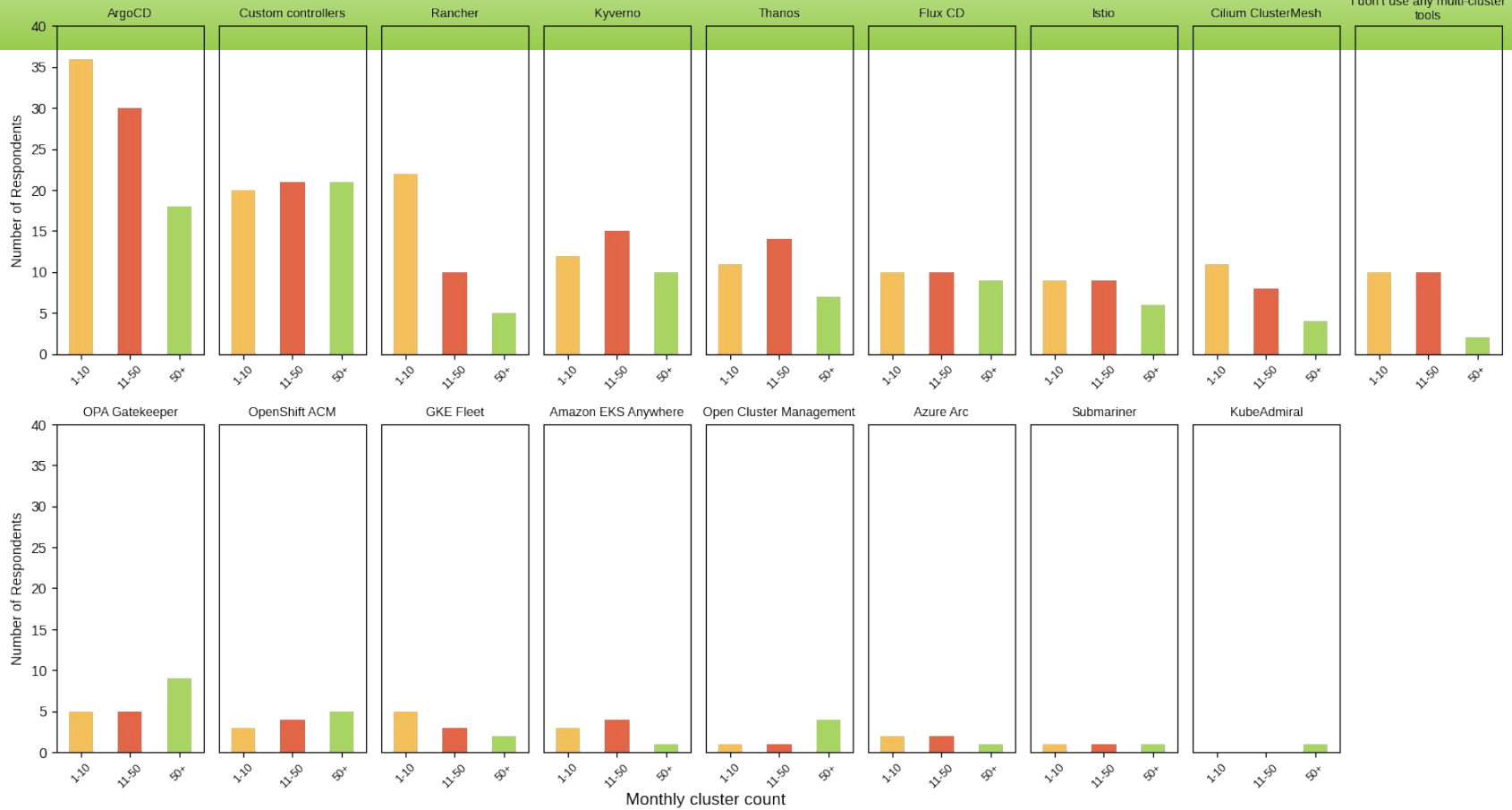
Tenancy Model of Clusters



Engineers Preferences on Automated Responses for Cross-Cluster Issues

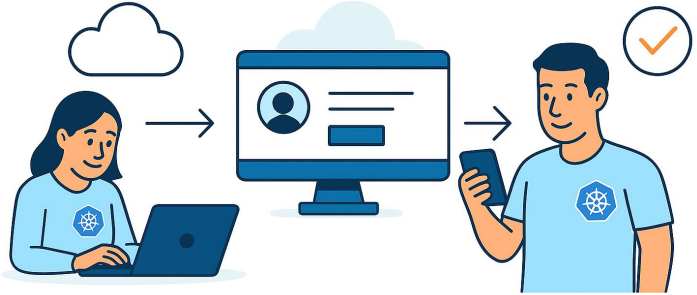


Quantitative highlights



User research

"Multiple clusters provide practical solutions to organizational, technical, and cost-related limitations that a single cluster cannot address."



- **Platform Engineer** (Manages 50+ Clusters)

User research Key Insights

Key Insights	Recommendations
1. Documentation gaps	➔ Provide clear up to date documentation with use cases.
2. Centralized monitoring	➔ Implement centralized observability solution.
3. AI-driven automation	➔ Leverage AI driven automation on tasks.
4. Operational complexity	➔ Reduce complexity, provide training.
5. Developer UX	➔ UI/UX & Developer Experience Improvement Plan.

"**Make it Easy**" - Software Developer

"Future Multi cluster Management = Technical Excellence + Human centered design."

Participate in the Survey

Multicluster challenges

Follow and get involved

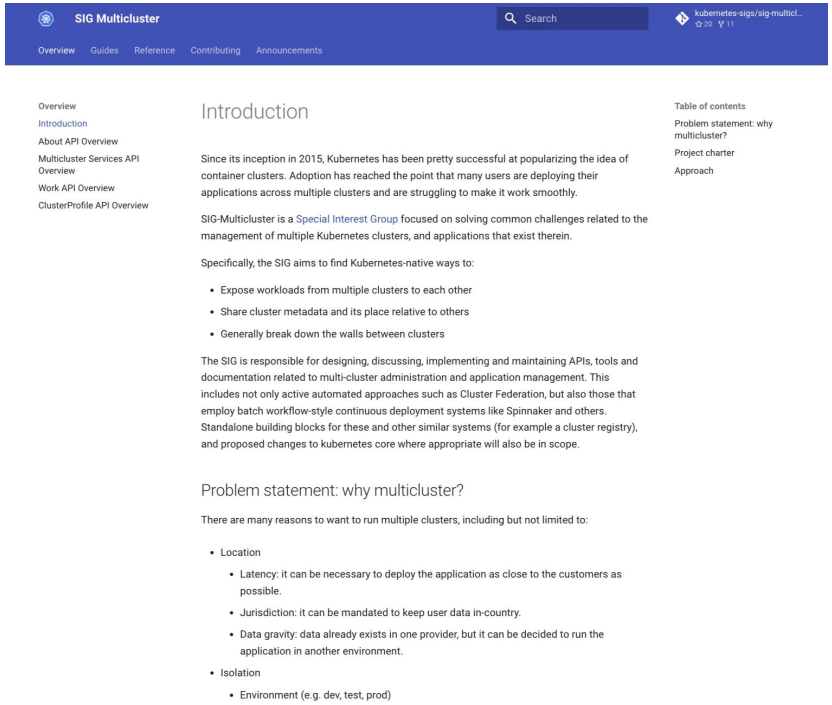
Be part of the SIG

- What are you working on?
  - APIs you're building, concepts you're missing — agenda is always open for demos!
- What problems do you or your customers have?
- Do you have unique needs that have been overlooked?
- Do you want to help?
  - Contribute to the test suite(s)
  - Add information to the web site
  - Join the UX team for more research and analysis
  - Stakeholder input on KEPs
  - Try out the credential providers

Be part of the SIG

Home page: https://multicluster.sigs.k8s.io
Slack channel:
https://kubernetes.slack.com/messages/sig-multicluster
List:
https://groups.google.com/forum/#!forum/kubernetes-sig-multicluster

Meetings are weekly Tuesdays, 12:30 Eastern, 9:30 Pacific, 16:30 UTC (AMER timeslot) and biweekly Thursdays, bi-weekly Thursday at 7 AM UTC (APAC/EMEA timeslot)



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Feedback: https://sched.co/27Nlj